

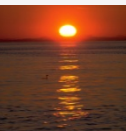
Parallel Databases





Parallel Systems

- Parallel database systems consist of multiple processors and multiple disks connected by a fast interconnection network.
- A **coarse-grain parallel** machine consists of a small number of powerful processors
- A **massively parallel** or **fine grain parallel** machine utilizes thousands of smaller processors.
- Two main performance measures:
 - **throughput** --- the number of tasks that can be completed in a given time interval
 - **response time** --- the amount of time it takes to complete a single task from the time it is submitted





Speed-Up and Scale-Up

- **Speedup**: a fixed-sized problem executing on a small system is given to a system which is N -times larger.

- Measured by:

$$\text{speedup} = \frac{\text{small system elapsed time}}{\text{large system elapsed time}}$$

- Speedup is **linear** if equation equals N .

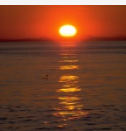
- **Scaleup**: increase the size of both the problem and the system

- N -times larger system used to perform N -times larger job

- Measured by:

$$\text{scaleup} = \frac{\text{small system small problem elapsed time}}{\text{big system big problem elapsed time}}$$

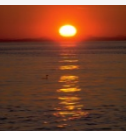
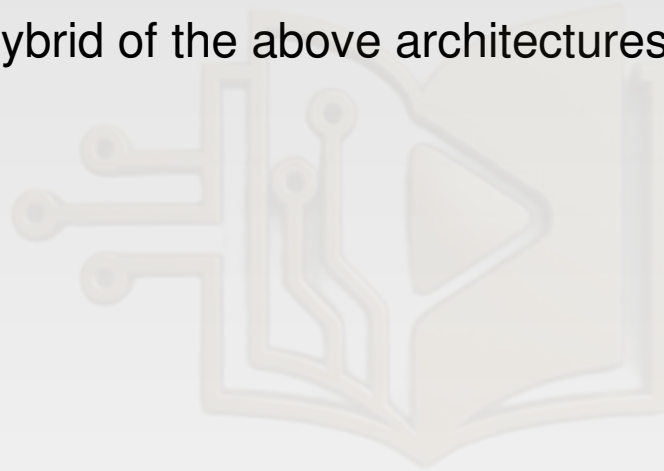
- Scale up is **linear** if equation equals 1.





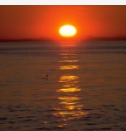
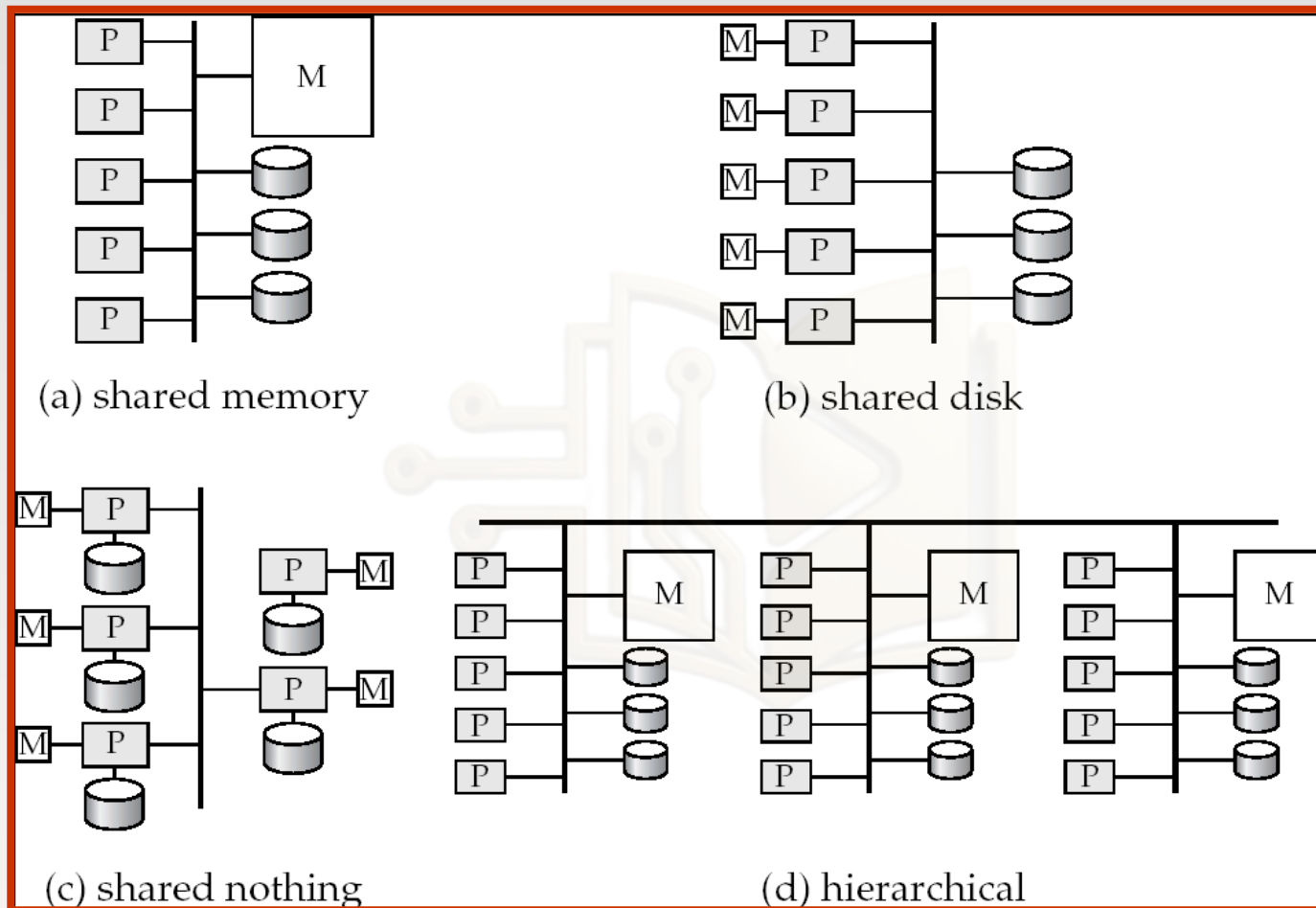
Parallel Database Architectures

- **Shared memory** -- processors share a common memory
- **Shared disk** -- processors share a common disk
- **Shared nothing** -- processors share neither a common memory nor common disk
- **Hierarchical** -- hybrid of the above architectures





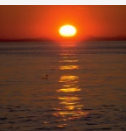
Parallel Database Architectures





Introduction

- **Parallel machines** are becoming quite common and affordable
 - **Prices** of microprocessors, memory and disks have dropped sharply
 - **Recent** desktop **computers** feature multiple processors and this trend is projected to accelerate
- **Databases** are growing increasingly large
 - **large volumes** of transaction **data** are collected and stored for later analysis.
 - **multimedia objects** like images are increasingly stored in databases
- **Large-scale** parallel **database** systems increasingly used for:
 - storing large volumes of data
 - processing time-consuming decision-support queries
 - providing high throughput for transaction processing





Parallelism in Databases

- Data can be **partitioned** across multiple disks for parallel I/O.
- **Individual** relational **operations** (e.g., sort, join, aggregation) can be executed in parallel
 - data can be partitioned and **each processor** can work independently on its own partition.
- **Queries** are expressed in high level language (SQL, translated to relational algebra)
 - makes **parallelization** easier.
- **Different queries** can be run in parallel with each other. Concurrency control takes care of conflicts.
- Thus, **databases** naturally **lend** themselves to parallelism.

